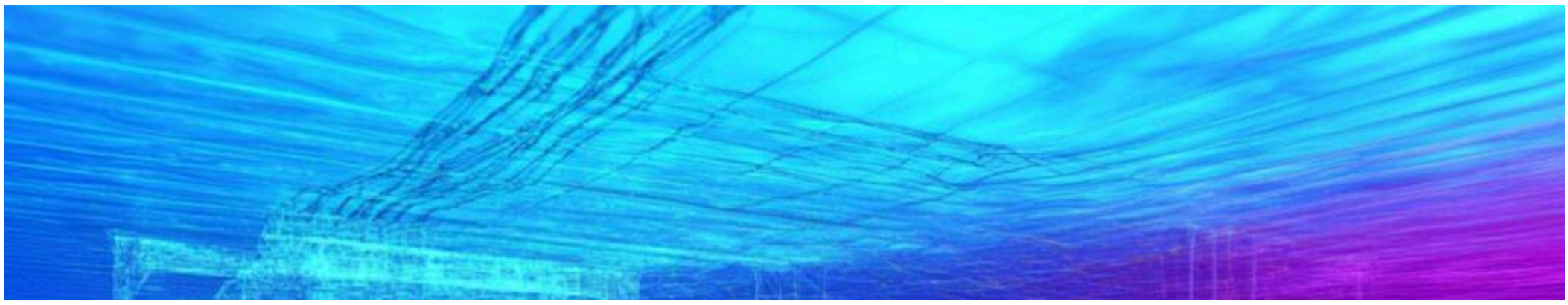

Two HEC Paris researchers, Olivier Chatain and Jerely Ghez, reveal how AI geopolitics is turning undersea cables into critical fault lines in a fragmented world economy.

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Key findings

- Submarine cables are now strategic infrastructure: These fiber-optic networks, powering 95% of global internet traffic, are central to geopolitical and industrial AI strategies.
- AI sovereignty is driving global fragmentation: The US, EU, and China are redefining digital borders with competing infrastructure.
- The risks of undersea disruption are rising,showing how vulnerable the system is.
- Tech giants are reshaping global connectivity: Firms like Google and Meta are investing in cable routes to avoid geopolitical chokepoints and reinforce network redundancy — often outpacing state action.

Audio reading

A- A+

From Shared Networks to Sovereign Ambitions

As the AI arms race ramps up tensions across the globe, the “geopolitical honeymoon” of the early days of the Internet is over, according to Jeremy Ghez and Olivier Chatain. In their ongoing research, funded by the HEC Foundation, they warn that this AI-driven “industrial revolution” could end in a “very messy separation”

Submarine Kingpins of the Global Economy

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between the US and China, centered on submarine cables. As the "great divorce" gathers speed, how can businesses adapt and leverage this new infrastructure?

From Shared Networks to Sovereign Ambitions

They're no thicker than a garden hose – albeit a hose encased in galvanized armor and clad in a polyethylene jacket. And yet these bundles of optical fibers buried beneath the ocean floor power up to 95% of all the data that zips around the globe. In all, there are 700,000 nautical miles of undersea cables. Even hidden from sight, there's no escaping Google and Meta (and, to a much lesser extent, Microsoft and Amazon), the heaviest investors in submarine cables. As for Elon Musk, his Starlink satellites simply can't match the massive amounts of data that cable can handle...

But how do these fiber optic cables stand up to the geopolitics of our planet? For ours is a tumultuous age. The trade showdown between the US and China, the ongoing wars in Sudan and Ukraine, the Israeli invasion of Gaza and the revolution in Syria are all stoking international divisions. And, looming over all this is the shadow of climate change.

> "In the past, technology united us, and interdependence was a source of stability.

These geopolitical frictions are shaping the AI revolution. "In the past, technology united us, and interdependence was a source of stability", explains Jeremy Ghez, the academic director of the HEC Center for Geopolitics, "but now these geopolitical rivalries have become technological rivalries". Governments worldwide are being driven by what Olivier Chatain, professor of competitive strategy at HEC, calls "political calculus". They are all bidding to be masters of their digital destiny. AI sovereignty is the name of this new game. Some entities – the US and the EU, for example – have already rolled out their own AI regulations and legislation to ringfence their national interests. If the trend is for control, the potential upshot is "global fragmentation, a patchwork of markets with little consistency, concedes Ghez.

Submarine Kingpins of the Global Economy

But let's come back to our underwater cables. Alongside profit-driven companies and the global telecom carriers, state actors rely on undersea networks for their communications. Although governments rarely own the infrastructure, they sometimes control routes indirectly via state-owned telecom operators. "In fact, the competition between rival states is fiercer than the competition between rival firms," claims Chatain.

Wrapped up in these undersea cables is a startling paradox: this "deep tech" is highly vulnerable. Sharks have been known to take a bite, ship anchors are a menace, and earthquakes pose a problem. Redundancy and swift repairs normally mitigate the impact of a single cut, but a series of incidents may together create prolonged disruptions as in South Africa in March 2024. More significantly, there is the threat of covert attacks stemming from the geopolitical tensions among Washington, Beijing, Russia and the EU. A series of cable cuts in the Baltic and Red Seas in late 2024 and around Taiwan in early 2025 raised eyebrows and have been investigated as instances of hostile actions below the threshold of war.

Governments monitor where new cables are installed. In 2020, the White House vetoed a planned subsea data cable between the US and Asia owned by Google and Meta and subsequently imposed a re-routing away from China out of concern for communication interceptions. Google is now developing networks in the Pacific and Indian ocean in an apparent bid to avoid the South China Sea. In late 2024, France took the decision to nationalize a strategic submarine network manufacturer and installer (Alcatel Submarine Networks) to prevent this "national security asset" from falling into the wrong hands.

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Double-edged Sword Beneath the Waves

This technological revolution is generating countless opportunities for a myriad of actors. “But not necessarily good actors,” points out Chatain. What would happen if cables are sabotaged in a coordinated way?” he asks. There would be little – or no – connectivity: national security communications would be compromised, financial transactions would grind to a halt, and critical services would hang in the balance. Interestingly, actors like Google, by investing massively in cable networks in out-of-the-way locations build redundancy and make the overall network more robust.

Both Chatain and Ghez agree it’s time to start thinking outside the (business) box. This new AI landscape is dominated not just by inter-firm rivalry, but also by rivalry – and conflict – between nations. As for civil society, it needs to ask itself if it wants to be a bystander, a pawn or an actor.

Read the highlights of the conference **"Strategy in an Age of Fragmentation".**

Sources

Article based on an interview with Olivier Chatain, professor of competitive strategy at HEC, and Jeremy Ghez, academic director of the HEC Center for Geopolitics, on their ongoing research, funded by the HEC Foundation.

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Olivier Chatain's main research focus is the formal and value-based foundations of strategy with specific applications to buyer-supplier relationships and markets for strategic resources.

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